The game is designed for 2-6 participants age 6 or more.

Game content:
72 atoms cards (of 5 different types)
36 Molecule cards (of 19 different types)

Instructions:

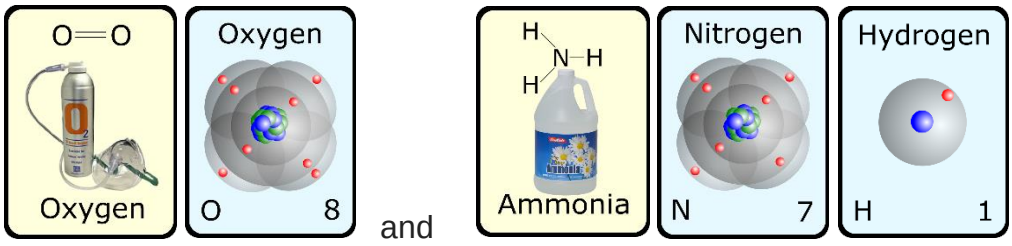
Game Preparation:
Shuffle the atoms cards and give 9 cards to each participant. Place the rest of the atoms cards in the stack that will be the “atoms’ deck”
Shuffle molecule cards and give 3 cards to each participant.
Place the rest of the molecules cards in the stack that will be the “molecules’ deck”

Purpose of the game:
To get rid of all the atoms cards. The first player to succeed is the winner.

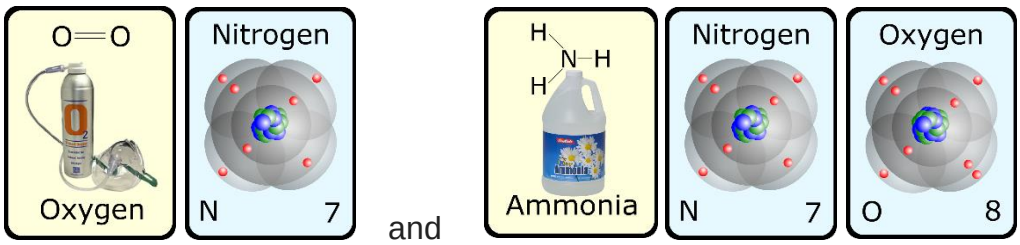
Gameplay:
The first player reveals one of his molecule cards and put it in the game area. He then lays one of his atom cards that appears in the molecule.

During the game, players in turn place cards of molecules and atoms in the game area according to the rules listed below.
The main rule is that at each stage, in the game area, there will be one molecule card and several atom cards so that together they form part of the molecule.

For example, the following situations are legal:



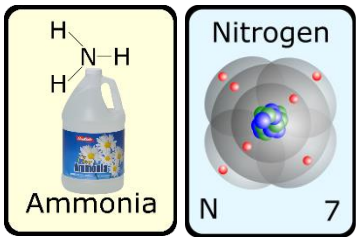
And the following situations are not:



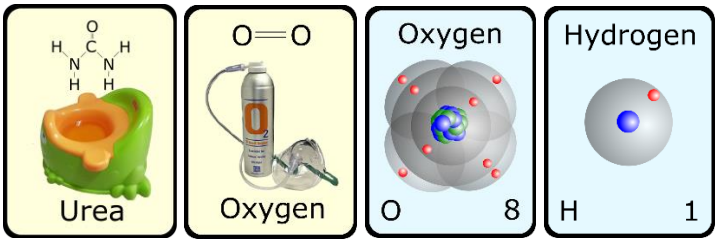
In turn, every player has three options:

- To put an atom card in the game area so that atoms placed in the game area together will form part of the molecule that in the molecule card that is placed in the game area.
- To discard (meaning to put into the molecule discard pile) the molecule card that currently in the game area, and to put his own molecule together with his own atom card on the game area such that all the cards in the game area together will form a part of the new molecule that he put in the game area.
- Take an atom card from the atoms' deck.

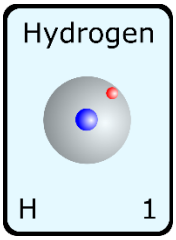
For example, if the following cards are lying in the game area,



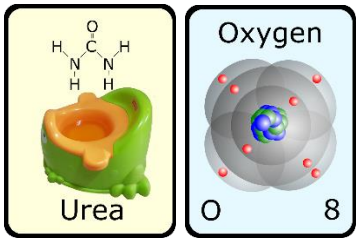
Then a player with the cards



Can either put an atom card:



Or alternatively to discard the molecule card that in the game area (in this case ammonia) and put a molecule card and an atom card:



<u>Finishing a molecule and starting an assembly of a new one:</u>

When a player completes the assembly of a molecule, he discards the molecule card. Then every player that participated in the molecule assembly can take one of the atom cards, which he has put, back to his hand. Then all the atoms cards from the game area are discarded and the player who finished forming the molecule has three options:

1. Start a new molecule in the same way as described earlier. The players will then continue to assemble the molecule as previously described.
2. Take 5 molecule cards from the molecules' deck and choose one of them for himself. The rest will be discarded.
3. Take 5 atom cards from the atom deck and choose one of them for himself. The rest will be discarded.

In case the player chooses options 2 or 3, the turn is passed to the next player and he can choose between these options, and so on.

Special Cards:
For beginner players, it is advisable to remove these cards from the stack and play without them first.

Alkanes card:
When a player places a card molecule it is written, "Alkenes" he should declare how many atoms of carbon have the alkan that is would like to construct. From that moment all will refer to this card as if it represents the alkane with that number of carbon atoms (and a corresponding number of hydrogen atoms: twice the number of Carbon atoms plus 2)

Nuclear fusion card:
A nuclear fusion card is an atom card. You can put this card together with a number (can also be 1) of atom cards. They will be considered as a single atom whose atomic number is equal to the sum of their atomic numbers. The atomic number is the number that appears at the bottom of the card.

Table with the atomic numbers of the Atoms that appear in the game:

Atom name	Symbol	Atomic number
Hydrogen	H	**1**
Carbon	C	**6**
Nitrogen	N	**7**
Oxygen	O	**8**

Appendices:
Remarks:
1. The game ends when one of the players get rid of all his atoms cards. There is no need to continue building the molecule (except if you would like to continue to play for the second place).
2. Each player sees only his own cards.
3. When one of the decks is finished, you shuffle the corresponding discard pile and continue playing (even when it is in the middle of molecule construction).
4. If both the atom deck and the atom discard pile are finished during the construction of a molecule then turn passes to the next player without the need to take an atom card. If a full circle is passed without any

player placed a card, then all the cards from the game area get discarded and the turn goes to the last plier that has put a card. Now he plays according to the rules described under “Finishing a molecule

Tips*:*

1. It is advisable to place the atom cards in the form in which they appear in the molecule card. This will make it easier to identify what atoms are missing.
2. Try to get rid of the Oxygen atoms. They are the hardest to get rid of.

Versions of the game:

Adjusting game length/number of people:

If there are many participants, or you are interested in the short game, then you can start the game with fewer atom cards for each participant; and vice versa - if there are few participants, or are interested in a longer game, then it is possible to start the game with more atom cards.
If you are interested in the short game, then you can start the game with more molecule cards for each participant.

Rummy/Rummikub Version:

Choose some number of molecule cards (you can do it randomly or not; the recommended number to start with is 5).
From here, you play only the atom cards and the rules of the game are the same as that of Rummy/Rummikub, only that instead of series you have to create molecules that appear on the cards of the selected molecules. The score of each atom card is its atomic number (the number that appears on the bottom of the card).

Simplified Rummy Version:

Choose some number of molecule cards and give the players only atom cards. Each player in his turn lays down several atom cards that form together one of the molecules that appear on the chosen molecule cards. If he cannot (or does not wish to) do it then he has to take an atom card from the deck. If the atoms’ deck runs out, you shuffle the used atom card and make a new atoms’ deck. The first player to get rid of all his cards is the winner.

Fast version:

Choose some number of molecule cards. Now you play as in the standard version, but without molecules cards: Instead, when it is your turn to start a molecule, you can choose any of the molecules that appear in the card that was chosen at the beginning of the game. The molecule card is not replaced until the molecule is finished, and is not discarded after the molecule is finished.